

SUJAL DHAKAL

(719) 373-7548 | hello@suajldhakal.dev | suajldhakal.dev | [linkedin.com/in/suajld](https://www.linkedin.com/in/suajld) | github.com/Sectonic | US Citizen

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science, GPA: 3.9/4.0

May 2027

- **Related Coursework:** Data Structures & Algorithms, Databases & SQL, Software Design & UML, Object-Oriented Programming, Computing in Python, Discrete Mathematics, Linear Algebra

EXPERIENCE

Bits of Good

Atlanta, GA

Software Developer

Jan. 2025 – Present

- Accelerated new feature development by **25%** by migrating the backend from raw MongoDB to Mongoose, **refactoring 10 services and 23 API endpoints** to enforce type safety and improve long-term code maintainability.
- Engineered full-stack medication management feature using Next.js and TailwindCSS, architecting 17 reusable React components and implementing schema-based validation with Zod, which **decreased user input errors by 60%**.
- Developed and deployed a **secure, token-based password reset system** using Node.js and Mongoose, integrating with an internal email API to automate the account recovery process and improve user security.

Technique Newspaper

Atlanta, GA

Software Developer

Nov. 2024 – Present

- **Slashed page load time by 80%** (from 10s to <2s) by architecting and deploying a new front-end for Georgia Tech's student newspaper using a headless Next.js system, leading to improved SEO performance and engagement.
- **Reduced backend load and cut API call redundancy by 55%** by engineering a custom service wrapper for the WordPress REST API, implementing an aggressive caching strategy that simplified data fetching.

Personality Academy

Remote

Co-Founder

Sept. 2022 – Aug. 2024

- Co-founded and **scaled an educational platform to over 3,000 users** by developing a full-stack application with Next.js, Supabase, and Prisma; implemented course delivery and a secure donation system via Stripe.
- Architected and built 'TypeSearch,' a **complex query tool for a database of 2,000+ individuals**, featuring multi-filter search logic and dynamic data visualization for 512 unique types.
- Engineered a **custom authentication system from the ground up**, integrating OAuth 2.0 for Google and Discord alongside secure email/password login, managing user data and sessions with Prisma ORM.

PROJECTS

Dementia Agitation Network | TypeScript, Python, React Native, Flask, Firebase, Spotify API, Expo, Git

[GitHub](#)

- **Secured \$1,000 in seed funding** and **won the 'Instructional Choice Award'** in a competitive 50-team showcase by pitching the project's technical architecture and market viability to investors and faculty.
- Led the development of the core mobile application in React Native, architecting a **real-time data pipeline from Apple Watch HealthKit to Firebase** to enable continuous monitoring of patient biometrics.
- Implemented an agitation detection model in Python that analyzed 5 biometric signals in real-time, **achieving 95% predictive accuracy** and automatically triggering therapeutic music interventions via the Spotify API.

Check-In | JavaScript, Next.js, PostgreSQL, Prisma, Recharts.js, Vercel, Git

[Website](#) | [GitHub](#)

- **Awarded 1st Place in the Congressional App Challenge (2022)** over 2,000+ competing teams nationwide, culminating in a presentation of the project's technical design at the U.S. Capitol.
- Designed and deployed a full-stack automated attendance system **serving over 1,000 students across 15+ organizations** and **reducing event check-in times by over 40 minutes**.
- Developed a real-time analytics dashboard with Recharts.js, featuring customizable data visualization and CSV export functionality, which **drove 100% adoption by administrators across an entire high school district**.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, HTML/CSS, Java, C#, C++, PHP, SQL

Libraries: Pandas, NumPy, Matplotlib, TensorFlow, Axios, Zod, D3.js, Jinja2, Mongoose

Frameworks: Flask, Django, React, React Native, NextJS, NodeJS, ExpressJS, Expo, Tailwind CSS, Bootstrap

Technologies: Git, GitHub, Google Cloud, Firebase, Vercel, Netlify, Docker, Stripe, Postman, MongoDB, Socket.IO